

# Gene Regulatory Network Construction: A Survey of Methods, Evidence, and Open Problems

Paperclip–Codex survey workflow

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## Abstract

Gene regulatory network (GRN) construction seeks to infer how transcription factors, target genes, cis-regulatory regions, perturbations, and cellular contexts jointly shape gene-expression programs. The field has moved from microarray-era reverse engineering and relevance networks to single-cell, multi-omic, perturbational, and representation-learning workflows. This survey synthesizes 73 evidence cards selected from 2,283 deduplicated candidates recovered through three rounds of Paperclip retrieval and Codex built-in web-search triangulation. The main conclusion is that GRN construction is not a single algorithmic problem. It is an evidence-integration problem in which the meaning of an edge changes with the data: co-expression, mutual information, conditional dependence, predictive importance, motif-supported TF-target relations, enhancer-mediated TF-region-gene circuits, dynamic influence, and perturbation-supported transcriptional responses or causal hypotheses are not interchangeable [28] [73] [41] [27] [1] [10] [15]. Classical methods remain important because they define the assumptions still visible in current tools: Bayesian and dynamic Bayesian models encode probabilistic structure [17] [49], information-theoretic methods rank dependence [41] [16] [46] [8], regression and tree ensembles produce regulator-to-target predictions [4] [27] [24] [47], and benchmarks demonstrate persistent data- and metric-dependence [39] [40] [51]. Recent directions add chromatin accessibility, motif priors, curated regulatory resources, perturbation design, eQTLs, spatial context, and deep representation learning [10] [70] [9] [63] [22] [7] [19] [69] [33] [34] [53]. High-quality GRN construction therefore requires explicit edge semantics, context-aware priors, benchmark caution, and validation strategies matched to the scientific question.

Figure 1 summarizes the survey’s central organizing claim: GRN methods should be interpreted by evidence strength, data richness, edge semantics, and cross-cutting validation constraints.

## 1 Introduction

A gene regulatory network is a model of regulatory relationships among molecular actors that influence gene expression. In a narrow transcriptional sense, its nodes are transcription factors and genes, and its directed edges indicate regulatory influence. In current workflows, GRN and GRN-adjacent models often include regulatory regions, chromatin accessibility, perturbations, cell-type-specific activity states, and, in intercellular extensions, ligand/receptor context [28] [10] [15] [5]. This expansion is scientifically valuable, but it also creates semantic ambiguity. An edge inferred from pairwise correlation, a GENIE3 feature-importance score, a SCENIC regulon, a SCENIC+ enhancer-mediated eRegulon, and a Perturb-seq response all suggest regulation, yet they carry different evidential commitments [27] [1] [10] [15].

The difficulty of GRN construction follows from a basic inverse problem. Regulatory edges are rarely observed directly. They are inferred from expression variation, time ordering, chromatin state, sequence motifs, curated knowledge, genetic variation, perturbational response, or some combination of these sources. Reviews across eras emphasize that the problem is high-dimensional, underdetermined, assumption-sensitive, and constrained by experimental design

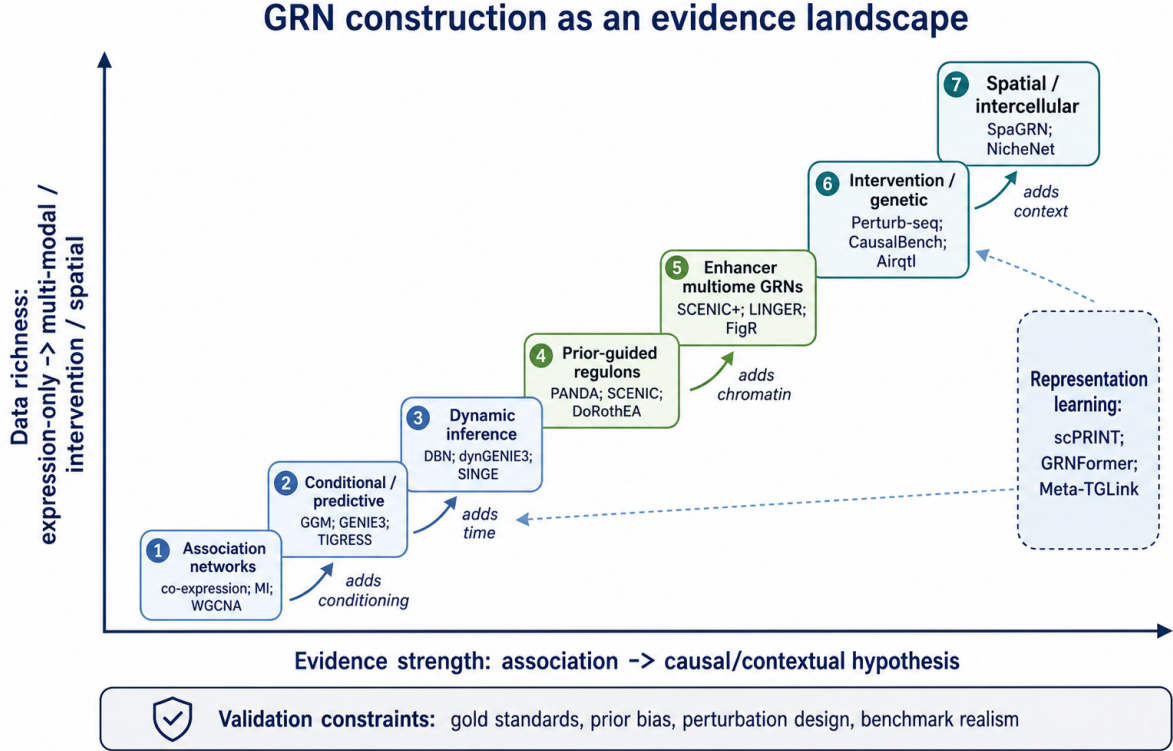


Figure 1: Evidence landscape for GRN construction.

[28] [73] [43] [60] [61]. A method can therefore fail either because the algorithm is weak or because the available data do not contain enough information to recover the target network. Ensemble and inferability analyses made this point explicit by showing that recoverability varies across networks and data regimes [40] [38].

The field’s terminology also reflects its history. Older papers used reverse engineering, gene-network inference, cellular network inference, relevance networks, and transcriptional regulatory network reconstruction [60] [42] [6]. Single-cell and multi-omic papers increasingly use regulon, TF activity, enhancer-mediated GRN, eRegulon, TF-region-gene circuit, and cell-state-specific GRN [28] [1] [10] [66]. This diversity is not merely lexical. It marks a shift from expression-only association graphs toward multi-evidence models that encode molecular plausibility and context. The purpose of this survey is to organize that shift without erasing the older assumptions that still determine what inferred networks mean.

## 2 Methods and Search Strategy

This review used a run-specific, auditable workflow. Paperclip searches covered broad umbrella terms, exact user terms, historical reverse-engineering terminology, named algorithms, benchmark resources, single-cell GRN methods, multi-omics, perturbational assays, eQTL methods, spatial transcriptomics, non-TF regulatory layers, and critique terms. Three Paperclip rounds produced 144 raw CSV exports and 5,191 raw rows. DOI/title merging and curated web integration yielded 2,283 deduplicated candidate records.

Codex built-in web search was used as a triangulation and metadata-verification layer rather than as a substitute for evidence extraction. Web search expanded retrieval around regulons, eRegulons, enhancer-driven GRNs, cell-type-specific networks, single-cell multiome data, CausalBench, Airqtl, SpaGRN, scPRINT, GRNFormer, and Meta-TGLink. It also corrected publication status: CausalBench, initially recorded as an adjacent preprint, was updated to its 2025

Communications Biology publication [9]. Web search also confirmed peer-reviewed metadata for SCENIC+, LINGER, GEARS, Geneformer, scGPT, scPRINT, Airqtl, SpaGRN, and Meta-TGLink [10] [70] [54] [25] [11] [33] [34] [53] [48].

Screening used four tiers: core, adjacent, context, and out\_of\_scope. Preprints were not excluded for being preprints, and candidate papers were kept in core, adjacent, or context unless a concrete topical mismatch was identified. Because the deduplicated pool exceeded 150 records, this run exceeded the minimum evidence threshold and created 73 evidence cards. The evidence cards support all citation-bearing claims in the manuscript and record publication status, evidence scope, method contribution, technical details, limitations, and supported claims; practical overview papers were used to check that the taxonomy remained useful rather than purely historical [45].

### 3 A Taxonomy of GRN Construction

The most important distinction is between association networks and regulatory mechanisms. Relevance networks and WGCNA-style co-expression modules are valuable for exploratory biology, module discovery, and hub-gene prioritization, but they generally produce undirected associations rather than direct TF-target regulation [6] [72]. Graphical Gaussian models and partial-correlation methods try to improve on pairwise association by estimating conditional dependence, and conservative methods such as C3NET prioritize sparse high-confidence edges [58] [2]. These methods can estimate conditional-dependence structure or select sparse high-confidence associations, yet they still do not by themselves establish biochemical direction or causality.

Information-theoretic GRN inference built a durable bridge between co-expression and regulatory-network construction. ARACNE estimates mutual information and prunes many indirect edges with the data processing inequality; CLR normalizes mutual information against gene-specific background distributions; MRNET uses maximum relevance/minimum redundancy feature selection; PIDC adapts information-theoretic reasoning to multivariate single-cell dependence [41] [16] [46] [8]. Their strength is that they can capture dependence beyond simple linear correlation and scale to genome-sized matrices. Their limitation is equally important: dependence is evidence for a candidate relationship, not proof of a direct regulatory edge.

Probabilistic and dynamical methods make stronger structural commitments. Bayesian networks introduced directed graphical models for expression data, while dynamic Bayesian networks added dependencies across time slices [17] [49]. Time-series reviews show why temporal ordering can help distinguish potential regulator and target, but only under sufficient sampling, synchronization, and model adequacy [43] [60]. Directed partial correlation, dynGENIE3, SCODE, SINCERITIES, Scribe, TENET, and SINGE illustrate different ways of importing dynamics, transfer entropy, Granger-style assumptions, or pseudotime into GRN inference [26] [18] [44] [52] [71] [31] [13]. These approaches support stronger temporal interpretations than static association methods, but they are sensitive to ordering, sampling density, hidden variables, and identifiability.

Regression, sparse feature selection, and tree ensembles remain central because they are scalable and produce target-wise regulator rankings. Inferelator integrates prior regulatory knowledge and systems-biology data; TIGRESS combines least-angle regression with stability selection; GENIE3 decomposes network inference into target-gene prediction with tree ensembles; GRNBoost2 improves scalability; NIMEFI and broader supervised/semi-supervised/unsupervised comparisons show how ensemble feature-importance and learning settings diversify this family; Inferelator 3.0 adapts prior-informed regression to large single-cell settings [4] [27] [24] [47] [37] [55] [65]. These methods are often practical and benchmark-competitive, but a regulator that predicts target expression is not necessarily the direct cause of that expression. Predictive edge semantics must therefore be reported explicitly.

A fourth family integrates prior and multi-modal evidence. PANDA combines motif-derived TF-target priors, protein-protein interaction information, and expression-derived co-regulation through message passing [69]. SCENIC combines co-expression modules, motif enrichment, and AUCell regulon-activity scoring to summarize single-cell regulatory programs [1]. SCENIC+, scMEGA, LINGER, FigR, and Pando extend this logic by using chromatin accessibility, motif evidence, region-to-gene links, paired multiome data, or external atlas-scale data to infer enhancer-mediated TF-region-gene relationships [10] [35] [70] [3] [68]. Cicero supplies a related cis-regulatory layer by predicting co-accessible DNA interactions from single-cell chromatin accessibility [64]. These methods enrich edge meaning, but they also add uncertainty: motif matches are not binding events, accessibility is not necessarily enhancer activity, and peak-to-gene links are often ambiguous [7] [64].

A fifth family uses interventions, genetic variation, or spatial context to move closer to causal interpretation. Early perturbational expression profiling already showed that interventions could constrain genetic-network hypotheses beyond passive observation [20]. Perturb-seq and CROP-seq later connected targeted perturbations with single-cell transcriptome-wide readouts, while TAP-seq and genome-scale Perturb-seq expanded the experimental design space [15] [12] [57] [21]. Perturbation-design studies show that using intervention labels explicitly can improve inference and that perturbation samples should not be treated as ordinary observations [59]. CausalBench now provides a peer-reviewed large-scale benchmark for network inference from single-cell perturbation data [9]. FINDR and Airqtl represent a separate genetics-based route, using eQTL or single-cell eQTL variation as causal information under suitable genetic-instrument and confounding assumptions [67] [33]. SpaGRN extends GRN construction into spatial transcriptomics by modeling spatially constrained receptor-TF-target regulons [34], while NicheNet borders GRN inference by linking ligands to target-gene programs through intercellular communication priors [5].

## 4 Comparative Evidence Across Method Families

The historical arc begins with expression association, but expression association should not be confused with direct regulation. Butte and Kohane’s relevance networks and WGCNA made network language useful for expression data by identifying gene clusters, modules, and hubs [6] [72]. Graphical Gaussian and partial-correlation approaches then asked whether conditioning on other variables could distinguish more direct associations [58] [71]. These methods are still useful as exploratory baselines, especially when the scientific aim is module discovery or candidate prioritization, but they support weaker biological claims than TF-target or perturbation-supported edges.

Information-theoretic methods sharpened association-based inference by focusing on dependence structure. ARACNE, CLR, and MRNET differ in pruning, normalization, and feature-selection logic, yet all address the same problem: dense expression dependence produces many false-positive or indirect candidate edges [41] [16] [46]. PIDC later showed that multivariate information measures could be adapted to single-cell data, where dropout and cellular heterogeneity complicate pairwise inference [8]. The conceptual lesson is durable: one can improve candidate-edge quality without changing the evidential category of the edge. The result remains association evidence unless additional temporal, motif, chromatin, genetic, or perturbational information is added.

Probabilistic and dynamical methods make the edge language more ambitious. Bayesian networks, dynamic Bayesian networks, directed partial correlation, dynGENIE3, and SINGE each introduce a way to talk about direction or temporal influence [17] [49] [26] [71] [13]. However, stronger language comes with stronger assumptions. Dynamic models require adequate sampling, pseudotime methods require reliable ordering, and Granger-style approaches require that predictive temporal precedence is meaningful in the biological system under study [43] [31]

[13]. For this reason, dynamic GRNs are best interpreted as model-based causal hypotheses rather than complete regulatory mechanisms.

Regression and tree-ensemble methods are influential because they scale, perform well in comparative settings, and match a natural biological question: which candidate regulators explain each target gene? GENIE3 became a benchmark reference point and an input to SCENIC-style workflows [27] [1]. GRNBoost2 preserved the target-wise regression logic while improving practical scalability [47]. TIGRESS and Inferelator represent stability-selection and prior-informed traditions [4] [24], and Inferelator 3.0 shows that this tradition remains active in single-cell-scale settings [65]. Yet regression edges are predictive relations. They become more biologically interpretable when the candidate regulator set is restricted to transcription factors, when prior evidence supports TF-target plausibility, or when chromatin and perturbation evidence agree.

Benchmarks have repeatedly disciplined the field’s claims. DREAM and DREAM5 demonstrated that no single inference strategy dominates across all networks and that aggregate community predictions can be more robust than individual methods [39] [40]. GeneNetWeaver and SERGIO made controlled in silico benchmarking possible, but also made simulator assumptions part of the evaluation environment [56] [14]. BEELINE brought this comparative logic to single-cell GRN inference and showed that performance depends on input data, metrics, and gold-standard definitions [51]. Benchmark realism remains difficult because curated regulatory resources such as RegulonDB and TRRUST are incomplete, organism-specific, and literature-biased [63] [22]. Thus benchmark results are necessary but not sufficient evidence of biological validity.

Single-cell data changed both opportunity and risk. SCENIC’s enduring contribution is not merely its edge list but its regulon-activity abstraction: it turns many candidate TF-target relations into cell-level activity programs that can be compared across cell states [1]. PIDC, SINCERITIES, SCODE, Scribe, LEAP, TENET, and SINGE show that single-cell variation can be exploited through information theory, time stamps, pseudotime, or dynamics [8] [18] [44] [52] [62] [31] [13]. At the same time, single-cell sparsity, dropout, cell-type mixture, and weak gold standards can produce unstable edges [51] [29]. Single-cell resolution therefore makes GRNs more context-specific, but it does not automatically make them more causal.

Multi-omics changes the standard of evidence by adding molecular plausibility. In expression-only inference, a TF-target edge may mean that TF expression predicts target expression. In enhancer-mediated multi-omic inference, the same visual edge may summarize TF expression, chromatin accessibility, motif evidence, region-to-gene association, and cell-type specificity [10] [35] [70] [3] [68]. Curated resources such as JASPAR, DoRothEA, RegulonDB, and TRRUST supply motifs, regulons, and prior interactions that help constrain the search space [63] [22] [7] [19]. The richer evidence improves interpretability, but it also multiplies assumptions. A rigorous report should state whether an edge is TF-gene, TF-region-gene, region-gene, regulon activity, or perturbation response.

Perturbational and genetic data provide stronger causal leverage than observational expression alone, but they do not eliminate modeling. Perturb-seq, CROP-seq, TAP-seq, and genome-scale Perturb-seq provide measured transcriptomic consequences of interventions [15] [12] [57] [21]. CellOracle uses inferred networks for in silico gene perturbation, and GEARS predicts transcriptional outcomes of novel multigene perturbations [30] [54]. CausalBench evaluates network inference from single-cell perturbation data, while Secilmis and colleagues show that knowledge of perturbation design itself matters [59] [9]. Genetic-variation methods such as FINDR and Airqtl provide complementary causal leverage through eQTLs [67] [33]. Yet perturbation effects include direct regulation, indirect cascades, feedback, cell-state shifts, and off-target effects, so intervention evidence must be interpreted with the experimental design.

Recent deep-learning and foundation-model methods add representation power rather than a settled replacement for GRN inference. GEARS, Geneformer, scGPT, and scFoundation show how large models or graph-enhanced models can predict perturbation responses, transfer

biological representations, or integrate large single-cell corpora [54] [25] [11] [23]. scPRINT is more directly framed as a large cell model for gene-network prediction, while GRNFormer and Meta-TGLink formulate GRN inference with graph transformers, graph autoencoding, or few-shot meta-learning [53] [36] [48]. These approaches are promising, particularly when data are sparse or labels are limited, but embeddings, attention, and predicted links require the same evidential scrutiny as older network scores.

## 5 Main Findings

First, GRN construction should be reported as an evidence-strength spectrum. Co-expression, relevance networks, mutual information, conditional dependence, regression feature importance, motif-supported regulons, enhancer-mediated circuits, perturbation responses, and eQTL-supported causal inference under instrument assumptions all occupy different positions on that spectrum [6] [58] [41] [27] [10] [15] [67]. A network diagram that fails to distinguish these categories invites overinterpretation.

Second, method choice should follow the data regime. Static bulk expression can support co-expression, information-theoretic, regression, and tree-based inference, but it is weak for causal direction [41] [27] [72]. Time-series or pseudotime data can support dynamic hypotheses if ordering and sampling are adequate [43] [26] [13]. scRNA-seq supports cell-state-specific regulon analysis but requires caution about sparsity and benchmark gold standards [1] [51] [29]. Paired scRNA-seq and scATAC-seq support enhancer-mediated GRNs, and perturbational or eQTL data support stronger causal hypotheses when perturbation design or genetic-instrument assumptions are explicit [10] [70] [15] [59] [67] [33].

Third, priors are often central but should be treated as evidence with uncertainty. RegulonDB and TRRUST provide curated regulatory interactions; JASPAR provides transcription-factor binding profiles; DoRothEA supports TF-activity estimation and regulon use [63] [22] [7] [19]. PANDA, SCENIC, SCENIC+, LINGER, and related methods demonstrate the value of prior-guided inference [1] [10] [70] [69]. However, priors are incomplete, literature-biased, organism-biased, and not automatically cell-state-specific. Confidence-aware use is preferable to unreported hard filtering.

Fourth, evaluation remains a central bottleneck. DREAM, DREAM5, GeneNetWeaver, SERGIO, BEELINE, and CausalBench collectively show that performance depends on network topology, simulator realism, perturbation coverage, gold-standard completeness, and metrics such as AUROC and AUPR [73] [39] [40] [56] [51] [14] [9]. The right question is not simply which method is best, but which method is best for a given edge meaning, data regime, and validation target.

Fifth, the field is broadening at its boundaries. miRNA-mediated networks remind us that transcription-factor-centered GRNs are not the only regulatory networks [50]. NicheNet and SpaGRN connect intracellular GRNs to intercellular signaling and spatial tissue context [34] [5]. scPRINT, GRNFormer, and Meta-TGLink are recent examples of representation-learning and graph-transfer components being introduced into GRN workflows [53] [36] [48]. These directions should be integrated cautiously, because prediction and representation are not the same as experimentally validated regulation.

## 6 Limitations of the Evidence Base

This survey is broad and representative rather than exhaustive. It includes landmark methods, benchmark papers, resource papers, single-cell and multi-omic workflows, perturbational and eQTL approaches, spatial/intercellular extensions, and selected recent deep-learning methods. It does not enumerate every organism-specific application or every incremental variant. That choice is appropriate for a general survey, but it means that specialized developmental, plant,

microbial, cancer-specific, and enhancer-biophysics literatures are represented mainly through methods and exemplars.

The evidence base also has a publication-status gradient. Most cited landmark and modern methods are peer-reviewed. Recent 2025-2026 work such as Airqtl, SpaGRN, scPRINT, GRNFormer, and Meta-TGLink was included because web search and Paperclip retrieval identified peer-reviewed or publisher-indexed versions by the current run date [33] [34] [53] [36] [48]. These papers are still young, so claims about their long-term impact are deliberately hedged.

Finally, GRN inference itself has unresolved validation problems. Gold standards and priors are incomplete, organism-specific, literature-biased, and limited by available motif, binding, expression, or perturbation evidence [63] [22] [7]. Perturbation data are powerful but can mix direct and indirect effects [59]. Single-cell data add dropout, sparsity, and cell-state heterogeneity [51] [29]. Multi-omic data add motif, accessibility, and peak-to-gene uncertainties [10] [64]. These are not merely limitations of this survey; they are central properties of the field.

## 7 Open Problems

The first open problem is edge semantics. Future GRN tools should report whether each edge represents co-expression, mutual information, conditional dependence, predictive importance, motif-supported TF-target plausibility, enhancer-mediated TF-region-gene evidence, dynamic influence, perturbation-supported effect, eQTL-supported causal inference under instrument assumptions, or spatial/intercellular regulatory context [6] [41] [58] [27] [10] [15] [67] [34]. Without this information, networks from different methods may look similar while meaning different things.

The second open problem is benchmark realism. Synthetic benchmarks provide known ground truth, but GeneNetWeaver and SERGIO encode assumptions about kinetics, topology, and noise [56] [14]. Curated regulatory resources provide biological evidence, but they are incomplete and biased [63] [22]. Perturbation benchmarks provide stronger intervention evidence, but they depend on perturbation design and response interpretation [59] [9]. A mature evaluation ecosystem will likely combine curated interactions, perturbation effects, chromatin evidence, time-series structure, uncertainty, and task-specific metrics.

The third open problem is context transfer. A GRN inferred in one organism, cell type, disease state, assay platform, or perturbation regime may not transfer to another. LIONESS, Inferelator 3.0, LINGER, Airqtl, SpaGRN, and Meta-TGLink each address parts of this problem by modeling sample-specific networks, large single-cell datasets, atlas-scale priors, cell-state-specific eQTLs, spatial regulons, or few-shot transfer [65] [32] [70] [33] [34] [48]. Together, these examples address context transfer through sample-specific networks, scalable single-cell inference, atlas-scale priors, cell-state eQTLs, spatial regulons, and few-shot transfer, but they do not yet provide a single settled transfer framework.

The fourth open problem is the relationship between prediction and explanation. Foundation models and graph neural networks can support perturbation-response modeling, regulatory prediction, gene-network prediction, or link prediction [54] [25] [11] [53] [36]. But a model that predicts expression well does not automatically provide a mechanistic GRN. Future work needs clearer protocols for converting representations into interpretable, falsifiable, and experimentally testable regulatory hypotheses.

## 8 Conclusion

Gene regulatory network construction has matured from a search for a single best inference algorithm into a multi-evidence modeling problem. Classical association, probabilistic, regression, and benchmark papers remain essential because they define the assumptions and evaluation failures that still shape modern tools [17] [41] [27] [39] [40]. Single-cell and multi-omic methods

add context, regulon activity, and enhancer-mediated evidence, while perturbational, eQTL, spatial, and foundation-model approaches expand what a GRN can represent [1] [10] [15] [33] [34] [53]. A defensible direction is not to replace older methods wholesale, but to make edge semantics explicit, combine evidence sources carefully, validate against the strongest available experimental or curated evidence, and state the limits of causal interpretation.

## References

- [1] Sara Aibar; Carmen Bravo Gonzalez-Blas; Thomas Moerman; Van Anh Huynh-Thu; Hana Imrichova; Gert Hulselmans; Florian Rambow; Jean-Christophe Marine; Pierre Geurts; Jan Aerts; Stein Aerts et al. Scenic: single-cell regulatory network inference and clustering. *Nature Methods*, 2017. doi: 10.1038/nmeth.4463. URL <https://www.nature.com/articles/nmeth.4463>. published peer-reviewed article.
- [2] Gokmen Altay and Frank Emmert-Streib. Inferring the conservative causal core of gene regulatory networks. *BMC Systems Biology*, 2010. doi: 10.1186/1752-0509-4-132. URL <https://doi.org/10.1186/1752-0509-4-132>. published peer-reviewed article.
- [3] Adam Kartha; Caleb J. B. et al. Figr: Functional inference of gene regulation using single-cell multi-omics. *Cell Genomics*, 2022. doi: 10.1016/j.xgen.2022.100166. URL <https://doi.org/10.1016/j.xgen.2022.100166>. published peer-reviewed article.
- [4] Richard Bonneau, David J. Reiss, Paul Shannon, Marc Facciotti, Leroy Hood, Nitin S. Baliga, and Vesteinn Thorsson. The inferelator: an algorithm for learning parsimonious regulatory networks from systems-biology data sets de novo. *Genome Biology*, 2006. doi: 10.1186/gb-2006-7-5-r36. URL <https://genomebiology.biomedcentral.com/articles/10.1186/gb-2006-7-5-r36>. published peer-reviewed article.
- [5] Robin Browaeys, Wouter Saelens, and Yvan Saeys. Nichenet: modeling intercellular communication by linking ligands to target genes. *Nature Methods*, 2020. doi: 10.1038/s41592-019-0667-5. URL <https://doi.org/10.1038/s41592-019-0667-5>. published peer-reviewed article.
- [6] Atul J. Butte and Isaac S. Kohane. Mutual information relevance networks: functional genomic clustering using pairwise entropy measurements. *Pacific Symposium on Biocomputing*, 2000. doi: 10.1142/9789814447331\\_0040. URL <https://pubmed.ncbi.nlm.nih.gov/10902190/>. published peer-reviewed article.
- [7] Castro-Mondragon et al. Jaspar 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 2024. doi: 10.1093/nar/gkad1059. URL <https://academic.oup.com/nar/article/52/D1/D174/7420101>. published peer-reviewed article.
- [8] Thalia E. Chan, Michael P. H. Stumpf, and Ann C. Babbie. Gene regulatory network inference from single-cell data using multivariate information measures. *Cell Systems*, 2017. doi: 10.1016/j.cels.2017.08.014. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC5624513/>. published peer-reviewed article.
- [9] Chevalley et al. A large-scale benchmark for network inference from single-cell perturbation data. *Communications Biology*, 2025. doi: 10.1038/s42003-025-07764-y. URL <https://doi.org/10.1038/s42003-025-07764-y>. published peer-reviewed article; earlier arXiv preprint located.

- [10] Carmen Bravo Gonzalez-Blas; Stijn De Winter; Gert Hulselmans; Nikolaus Hecker; Vlad Matetovici; Valerie Christiaens et al. Scenic+: single-cell multiomic inference of enhancers and gene regulatory networks. *Nature Methods*, 2023. doi: 10.1038/s41592-023-01938-4. URL <https://www.nature.com/articles/s41592-023-01938-4>. published peer-reviewed article.
- [11] Haotian Cui et al. scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nature Methods*, 2024. doi: 10.1038/s41592-024-02201-0. URL <https://www.nature.com/articles/s41592-024-02201-0>. published peer-reviewed article.
- [12] Paul Datlinger, Andre F. Rendeiro, Christoph Schmidl, Thomas Krausgruber, Peter Traxler, Johanna Klughammer, Linda C. Schuster, Amelie Kuchler, Donat Alpar, and Christoph Bock. Pooled crispr screening with single-cell transcriptome readout. *Nature Methods*, 2017. doi: 10.1038/nmeth.4177. URL <https://pubmed.ncbi.nlm.nih.gov/28099430/>. published peer-reviewed article.
- [13] Atul Deshpande, Li-Fang Chu, Ron Stewart, and Anthony Gitter. Network inference with granger causality ensembles on single-cell transcriptomics. *Cell Reports*, 2022. doi: 10.1016/j.celrep.2022.110333. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC9093087/>. published peer-reviewed article.
- [14] Payam Dibaeinia and Saurabh Sinha. Sergio: A single-cell expression simulator guided by gene regulatory networks. *Cell Systems*, 2020. doi: 10.1016/j.cels.2020.08.003. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC7530147/>. published peer-reviewed article.
- [15] Atray Dixit, Oren Parnas, Biyu Li, Jenny Chen, Charles P. Fulco, Livnat Jerby-Arnon, Nemanja Marjanovic, Danielle Dionne, Tyler Burks, Britt Adamson, Thomas M. Norman, Eric Lander, Jonathan Weissman, Nir Friedman, and Aviv Regev. Perturb-seq: Dissecting molecular circuits with scalable single-cell rna profiling of pooled genetic screens. *Cell*, 2016. doi: 10.1016/j.cell.2016.11.038. URL <https://www.ncbi.nlm.nih.gov/bioproject/354362>. published peer-reviewed article.
- [16] Jeremiah J. Faith, Boris Hayete, Joshua T. Thaden, Ilaria Mogno, Jamey Wierzbowski, Guillaume Cottarel, Simon Kasif, James J. Collins, and Timothy S. Gardner. Large-scale mapping and validation of escherichia coli transcriptional regulation from a compendium of expression profiles. *PLoS Biology*, 2007. doi: 10.1371/journal.pbio.0050008. URL <https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.0050008>. published peer-reviewed article.
- [17] Nir Friedman, Michal Linial, Iftach Nachman, and Dana Peer. Using bayesian networks to analyze expression data. *Journal of Computational Biology*, 2000. doi: 10.1089/106652700750050961. URL <https://doi.org/10.1089/106652700750050961>. published peer-reviewed article.
- [18] Nguyen Phong Papili Gao, S. M. M. Ud-Dean, Olivier Gandrillon, and Rudyanto Gunawan. Sincerities: inferring gene regulatory networks from time-stamped single cell transcriptional expression profiles. *Bioinformatics*, 2018. doi: 10.1093/bioinformatics/btx575. URL <https://academic.oup.com/bioinformatics/article/34/2/258/4158033>. published peer-reviewed article.
- [19] Garcia-Alonso et al. Benchmark and integration of resources for the estimation of human transcription factor activities. *Genome Research*, 2019. doi: 10.1101/gr.240663.118. URL <https://saezlab.github.io/dorothea/index.html>. published peer-reviewed article.

- [20] Timothy S. Gardner, Diego di Bernardo, David Lorenz, and James J. Collins. Inferring genetic networks and identifying compound mode of action via expression profiling. *Science*, 2003. doi: 10.1126/science.1081900. URL <https://doi.org/10.1126/science.1081900>. published peer-reviewed article.
- [21] Joseph M. Replogle; Reuben A. Saunders; Angela N. Pogson; Jeffrey A. Hussmann; Alex Lenail; Arjun Guna et al. Mapping information-rich genotype-phenotype landscapes with genome-scale perturb-seq. *Cell*, 2022. doi: 10.1016/j.cell.2022.05.013. URL <https://pubmed.ncbi.nlm.nih.gov/35688146/>. published peer-reviewed article.
- [22] Han et al. Trrust: a reference database of human transcriptional regulatory interactions. *Scientific Reports*, 2015. doi: 10.1038/srep11432. URL <https://www.nature.com/articles/srep11432>. published peer-reviewed article.
- [23] Hao et al. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 2024. doi: 10.1038/s41592-024-02305-7. URL <https://www.nature.com/articles/s41592-024-02305-7>. published peer-reviewed article.
- [24] Anne-Claire Haury, Fantine Mordelet, Paola Vera-Licona, and Jean-Philippe Vert. Tigrass: Trustful inference of gene regulation using stability selection. *BMC Systems Biology*, 2012. doi: 10.1186/1752-0509-6-145. URL <https://bmcsystbiol.biomedcentral.com/articles/10.1186/1752-0509-6-145>. published peer-reviewed article.
- [25] Christina V. Theodoris; Ling Xiao; Anant Chopra; Mark D. Chaffin; Ziad R. Al Sayed; Matthew C. Hill et al. Transfer learning enables predictions in network biology. *Nature*, 2023. doi: 10.1038/s41586-023-06139-9. URL <https://www.nature.com/articles/s41586-023-06139-9>. published peer-reviewed article.
- [26] Van Anh Huynh-Thu and Pierre Geurts. dyngenie3: dynamical genie3 for the inference of gene networks from time series expression data. *Scientific Reports*, 2018. doi: 10.1038/s41598-018-21715-0. URL <https://www.nature.com/articles/s41598-018-21715-0>. published peer-reviewed article.
- [27] Van Anh Huynh-Thu, Alexandre Irrthum, Louis Wehenkel, and Pierre Geurts. Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE*, 2010. doi: 10.1371/journal.pone.0012776. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0012776>. published peer-reviewed article.
- [28] Pau Badia i Mompel, Lorna Wessels, Sophia Muller-Dott, Remi Trimbou, Ricardo O. Ramirez Flores, Ricard Argelaguet, and Julio Saez-Rodriguez. Gene regulatory network inference in the era of single-cell multi-omics. *Nature Reviews Genetics*, 2023. doi: 10.1038/s41576-023-00618-5. URL <https://www.nature.com/articles/s41576-023-00618-5>. published peer-reviewed article.
- [29] Christopher A. Jackson et al. Evaluating methods of inferring gene regulatory networks highlights their lack of performance for single cell gene expression data. *BMC Bioinformatics*, 2018. doi: 10.1186/s12859-018-2217-z. URL <https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-018-2217-z>. published peer-reviewed article.
- [30] Kamimoto et al. Dissecting cell identity via network inference and in silico gene perturbation. *Nature*, 2023. doi: 10.1038/s41586-022-05688-9. URL <https://www.nature.com/articles/s41586-022-05688-9>. published peer-reviewed article.

- [31] Dylan Kim et al. Tenet: gene network reconstruction using transfer entropy reveals key regulatory factors from single cell transcriptomic data. *Nucleic Acids Research*, 2021. doi: 10.1093/nar/gkaa1014. URL <https://doi.org/10.1093/nar/gkaa1014>. published peer-reviewed article.
- [32] Marieke Lydia Kuijjer et al. Single-sample network inference with lioness. *iScience*, 2019. doi: 10.1016/j.isci.2019.03.021. URL <https://doi.org/10.1016/j.isci.2019.03.021>. published peer-reviewed article.
- [33] Kexin Li et al. Airqtl dissects cell state-specific causal gene regulatory networks with efficient single-cell eqtl mapping. *Nature Communications*, 2025. doi: 10.1038/s41467-025-66214-9. URL <https://www.nature.com/articles/s41467-025-66214-9>. published peer-reviewed article.
- [34] Yao Li, Xiaobin Liu, Lidong Guo, Kai Han, Shuangfang Fang, Xinjiang Wan, Xun Xu, Guangyi Fan, and Mengyang Xu. Spagr: Investigating spatially informed regulatory paths for spatially resolved transcriptomics data. *Cell Systems*, 2025. doi: 10.1016/j.cels.2025.101243. URL <https://doi.org/10.1016/j.cels.2025.101243>. published peer-reviewed article.
- [35] Zhijian Li, James S. Nagai, Christoph Kuppe, Rafael Kramann, and Ivan G. Costa. scmega: single-cell multi-omic enhancer-based gene regulatory network inference. *Bioinformatics Advances*, 2023. doi: 10.1093/bioadv/vbad003. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC9853317/>. published peer-reviewed article.
- [36] Kazi M. and collaborators. Grnformer: accurate gene regulatory network inference using graph transformer. *Bioinformatics*, 2026. doi: 10.1093/bioinformatics/btag144. URL <https://academic.oup.com/bioinformatics/article/42/4/btag144/8540455>. published peer-reviewed article.
- [37] Stefan R. Maetschke, Piyush B. Madhamshekar, Melissa J. Davis, and Mark A. Ragan. Supervised, semi-supervised and unsupervised inference of gene regulatory networks. *Briefings in Bioinformatics*, 2014. doi: 10.1093/bib/bbt034. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC3956069/>. published peer-reviewed article.
- [38] Marbach et al. Ensemble inference and inferability of gene regulatory networks. *PLoS ONE*, 2014. doi: 10.1371/journal.pone.0103812. URL <https://doi.org/10.1371/journal.pone.0103812>. published peer-reviewed article.
- [39] Daniel Marbach, Robert J. Prill, Thomas Schaffter, Claudio Mattiussi, Dario Floreano, and Gustavo Stolovitzky. Revealing strengths and weaknesses of methods for gene network inference. *PNAS*, 2010. doi: 10.1073/pnas.0913357107. URL <https://pubmed.ncbi.nlm.nih.gov/20308593/>. published peer-reviewed article.
- [40] Daniel Marbach, James C. Costello, Robert Kuffner, Nicole M. Vega, Robert J. Prill, Diogo M. Camacho, Kyle R. Allison, DREAM5 Consortium, Manolis Kellis, James J. Collins, and Gustavo Stolovitzky. Wisdom of crowds for robust gene network inference. *Nature Methods*, 2012. doi: 10.1038/nmeth.2016. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC3512113/>. published peer-reviewed article.
- [41] Adam A. Margolin, Ilya Nemenman, Katia Basso, Chris Wiggins, Gustavo Stolovitzky, Riccardo Dalla Favera, and Andrea Califano. Aracne: An algorithm for the reconstruction of gene regulatory networks in a mammalian cellular context. *BMC Bioinformatics*, 2006. doi: 10.1186/1471-2105-7-S1-S7. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC1810318/>. published peer-reviewed article.

- [42] Florian Markowetz and Rainer Spang. Inferring cellular networks - a review. *BMC Bioinformatics*, 2007. doi: 10.1186/1471-2105-8-S6-S5. URL <https://bmcbioinformatics.biomedcentral.com/articles/10.1186/1471-2105-8-S6-S5>. published peer-reviewed article.
- [43] Malvina Marku et al. From time-series transcriptomics to gene regulatory networks: A review on inference methods. *PLOS Computational Biology*, 2023. doi: 10.1371/journal.pcbi.1011254. URL <https://doi.org/10.1371/journal.pcbi.1011254>. published peer-reviewed article.
- [44] Hirotaka Matsumoto, Hisanori Kiryu, Chikara Furusawa, Minoru S. H. Ko, Shigeru B. H. Ko, Norio Gouda, Tetsutaro Hayashi, and Itoshi Nikaido. Scode: an efficient regulatory network inference algorithm from single-cell rna-seq during differentiation. *Bioinformatics*, 2017. doi: 10.1093/bioinformatics/btx194. URL <https://academic.oup.com/bioinformatics/article/33/15/2314/3100331>. published peer-reviewed article.
- [45] Mercatelli, Scalambra, Triboli, Ray, and Giorgi. Gene regulatory network inference resources: A practical overview. *BBA - Gene Regulatory Mechanisms*, 2020. doi: 10.1016/j.bbagr.2019.194430. URL <https://www.sciencedirect.com/science/article/abs/pii/S1874939919300410>. published peer-reviewed article.
- [46] Patrick E. Meyer, Frederic Lafitte, and Gianluca Bontempi. Information-theoretic inference of large transcriptional regulatory networks. *BMC Bioinformatics*, 2008. doi: 10.1186/1471-2105-9-461. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC2630331/>. published peer-reviewed article.
- [47] Thomas Moerman, Sara Aibar Santos, Carmen Bravo Gonzalez-Blas, Jana Simm, Yves Moreau, Jan Aerts, and Stein Aerts. Grnboost2 and arboreto: efficient and scalable inference of gene regulatory networks. *Bioinformatics*, 2019. doi: 10.1093/bioinformatics/bty916. URL <https://academic.oup.com/bioinformatics/article/35/12/2159/5184284>. published peer-reviewed article.
- [48] Weiming Yu; Zhuobin Chen; Yaohua Hu; Jing Qin; Le Ou-Yang et al. Structure-enhanced graph meta learning for few-shot gene regulatory network inference. *Genome Biology*, 2025. doi: 10.1186/s13059-025-03860-8. URL <https://genomebiology.biomedcentral.com/articles/10.1186/s13059-025-03860-8>. published peer-reviewed article.
- [49] Blaise Perrin, Liva Ralaivola, Aurelien Mazurie, Sebastien Bottani, Jean Mallet, and Florence d Alche Buc. Gene networks inference using dynamic bayesian networks. *Bioinformatics*, 2003. doi: 10.1093/bioinformatics/btg1071. URL <https://doi.org/10.1093/bioinformatics/btg1071>. published peer-reviewed article.
- [50] Inna P. Plaisier et al. Overview of computational approaches for inference of microrna-mediated and gene regulatory networks. *Methods in Molecular Biology*, 2013. doi: 10.1007/978-1-62703-514-9\_14. URL [https://doi.org/10.1007/978-1-62703-514-9\\_14](https://doi.org/10.1007/978-1-62703-514-9_14). published peer-reviewed article.
- [51] Aditya Pratapa, Amogh P. Jalihal, Jeffrey N. Law, Aditya Bharadwaj, and T. M. Murali. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 2020. doi: 10.1038/s41592-019-0690-6. URL <https://www.nature.com/articles/s41592-019-0690-6>. published peer-reviewed article.
- [52] Xiaojie Qiu et al. Inferring causal gene regulatory networks from coupled single-cell expression dynamics using scribe. *Cell Systems*, 2020. doi: 10.1016/j.cels.2020.02.003. URL <https://pubmed.ncbi.nlm.nih.gov/32135093/>. published peer-reviewed article.

- [53] Cédric Reuter et al. sprint: pre-training on 50 million cells allows robust gene network predictions. *Nature Communications*, 2025. doi: 10.1038/s41467-025-58699-1. URL <https://www.nature.com/articles/s41467-025-58699-1>. published peer-reviewed article.
- [54] Yusuf Roohani, Kexin Huang, and Jure Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with gears. *Nature Biotechnology*, 2024. doi: 10.1038/s41587-023-01905-6. URL <https://www.nature.com/articles/s41587-023-01905-6>. published peer-reviewed article.
- [55] Joeri Ruyssinck, Van Anh Huynh-Thu, Pierre Geurts, Tom Dhaene, Piet Demeester, and Yvan Saeys. Nimefi: Gene regulatory network inference using multiple ensemble feature importance algorithms. *PLoS ONE*, 2014. doi: 10.1371/journal.pone.0092709. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0092709>. published peer-reviewed article.
- [56] Thomas Schaffter, Daniel Marbach, and Dario Floreano. Genenetweaver: in silico benchmark generation and performance profiling of network inference methods. *Bioinformatics*, 2011. doi: 10.1093/bioinformatics/btr373. URL <https://academic.oup.com/bioinformatics/article/27/16/2263/254752>. published peer-reviewed article.
- [57] Daniel Schraivogel et al. Targeted perturb-seq enables genome-scale genetic screens in single cells. *Nature Methods*, 2020. doi: 10.1038/s41592-020-0837-5. URL <https://www.nature.com/articles/s41592-020-0837-5>. published peer-reviewed article.
- [58] Juliane Schäfer and Korbinian Strimmer. An empirical bayes approach to inferring large-scale gene association networks. *Bioinformatics*, 2005. doi: 10.1093/bioinformatics/bti062. URL <https://doi.org/10.1093/bioinformatics/bti062>. published peer-reviewed article.
- [59] Deniz Secilmis et al. Knowledge of the perturbation design is essential for accurate gene regulatory network inference. *Scientific Reports*, 2022. doi: 10.1038/s41598-022-19005-x. URL <https://doi.org/10.1038/s41598-022-19005-x>. published peer-reviewed article.
- [60] Chung S. Sima, Jing Hua, and Sunghoon Jung. Inference of gene regulatory networks using time-series data: A survey. *Current Genomics*, 2009. doi: 10.2174/138920209789177610. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC2766792/>. published peer-reviewed article.
- [61] Riet De Smet and Kathleen Marchal. Advantages and limitations of current network inference methods. *Nature Reviews Microbiology*, 2010. doi: 10.1038/nrmicro2419. URL <https://pubmed.ncbi.nlm.nih.gov/20805835/>. published peer-reviewed article.
- [62] Aaron T. Specht and Jun Li. Leap: constructing gene co-expression networks for single-cell rna-sequencing data using pseudotime ordering. *Bioinformatics*, 2017. doi: 10.1093/bioinformatics/btw729. URL <https://academic.oup.com/bioinformatics/article/33/5/764/2557687>. published peer-reviewed article.
- [63] Tierrafría et al. Regulondb v12.0: a comprehensive resource of transcriptional regulation in escherichia coli k-12. *Nucleic Acids Research*, 2024. doi: 10.1093/nar/gkad1072. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC10767902/>. published peer-reviewed article.
- [64] Hannah A. Pliner; Jay Shendure; Cole Trapnell et al. Cicero predicts cis-regulatory dna interactions from single-cell chromatin accessibility data. *Molecular Cell*, 2018. doi: 10.1016/j.molcel.2018.06.044. URL <https://doi.org/10.1016/j.molcel.2018.06.044>. published peer-reviewed article.

- [65] Claudia Skok Gibbs; Christopher A. Jackson; Giuseppe-Antonio Saldi; Andreas Tjarnberg; Aashna Shah; Aaron Watters; Nicholas De Veaux et al. High-performance single-cell gene regulatory network inference at scale: the inferelator 3.0. *Bioinformatics*, 2022. doi: 10.1093/bioinformatics/btac117. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC9048651/>. published peer-reviewed article.
- [66] Wang et al. Gene regulatory network reconstruction: harnessing the power of single-cell multi-omic data. *npj Systems Biology and Applications*, 2023. doi: 10.1038/s41540-023-00312-6. URL <https://www.nature.com/articles/s41540-023-00312-6>. published peer-reviewed article.
- [67] Lingfei Wang and Tom Michoel. Efficient and accurate causal inference with hidden confounders from genome-transcriptome variation data. *PLoS Computational Biology*, 2017. doi: 10.1371/journal.pcbi.1005703. URL <https://doi.org/10.1371/journal.pcbi.1005703>. published peer-reviewed article.
- [68] Jonas S. Fleck; Sara M. Jansen; Dominik Wollny et al. Inferring and perturbing cell fate regulomes in human brain organoids. *Nature*, 2023. doi: 10.1038/s41586-022-05279-8. URL <https://www.nature.com/articles/s41586-022-05279-8>. published peer-reviewed article.
- [69] Kimberly Glass; Curtis Huttenhower; John Quackenbush; Guo-Cheng Yuan et al. Passing messages between biological networks to refine predicted interactions. *PLoS ONE*, 2013. doi: 10.1371/journal.pone.0064832. URL <https://doi.org/10.1371/journal.pone.0064832>. published peer-reviewed article.
- [70] Qiuyue Yuan and Zhana Duren. Inferring gene regulatory networks from single-cell multiome data using atlas-scale external data. *Nature Biotechnology*, 2025. doi: 10.1038/s41587-024-02182-7. URL <https://www.nature.com/articles/s41587-024-02182-7>. published peer-reviewed article.
- [71] Yinyin Yuan, Chang-Tsun Li, and Oliver Windram. Directed partial correlation: Inferring large-scale gene regulatory network through induced topology disruptions. *PLoS ONE*, 2011. doi: 10.1371/journal.pone.0016835. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0016835>. published peer-reviewed article.
- [72] Bin Zhang and Steve Horvath. A general framework for weighted gene co-expression network analysis. *Statistical Applications in Genetics and Molecular Biology*, 2005. doi: 10.2202/1544-6115.1128. URL <https://doi.org/10.2202/1544-6115.1128>. published peer-reviewed article.
- [73] Mengyuan Zhao et al. A comprehensive overview and critical evaluation of gene regulatory network inference technologies. *Briefings in Bioinformatics*, 2021. doi: 10.1093/bib/bbab009. URL <https://academic.oup.com/bib/article/22/5/bbab009/6128842>. published peer-reviewed article.